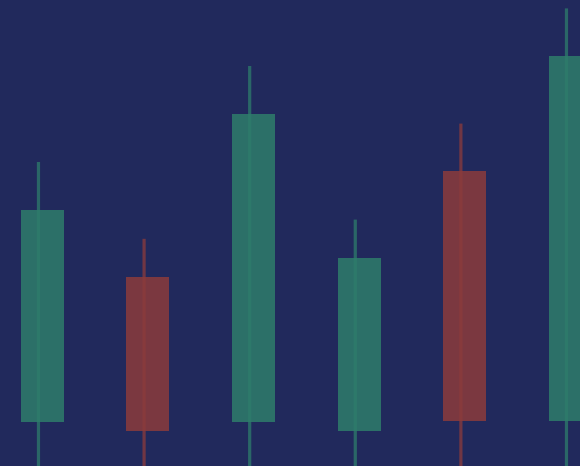


Stock Price Prediction

Using Machine Learning & Time-Series Analysis

NAME	Prince Kumar
ROLL NO.	25SCS1003002796
COLLEGE	IILM University, Greater Noida
SEMESTER	3rd Semester



What Is Stock Price Prediction?

Stock price prediction is the process of forecasting the future value of a company's stock using historical price data, trading volume, and other market indicators. It combines finance, statistics, and machine learning to identify patterns that may indicate future price movement.



Why it matters

Helps investors, analysts and institutions make informed, data-driven decisions.



Nature of the problem

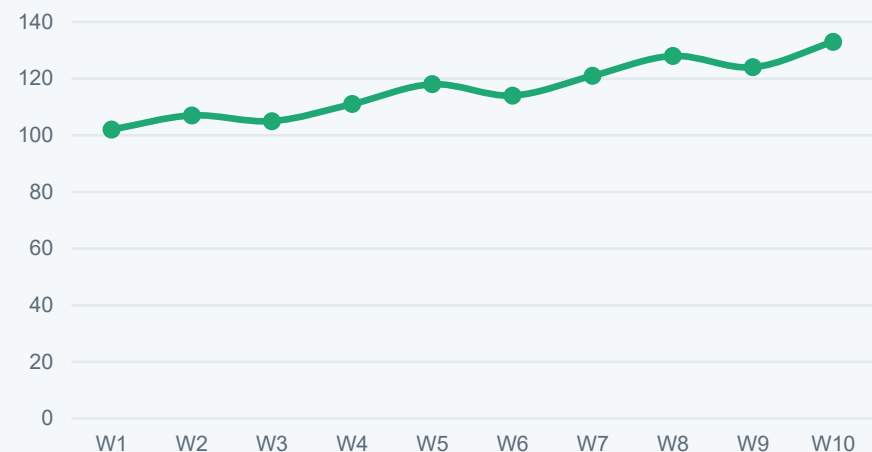
Stock markets are highly volatile, non-linear and influenced by many external factors.



Our approach

Use historical OHLC data with ML / deep-learning models to forecast future closing prices.

SAMPLE CLOSING PRICE TREND



Real-world prices form noisy, non-linear time series — the pattern ML models are trained to learn.

The Challenge We Are Addressing

Traditional analysis methods struggle to capture the complex, non-linear behaviour of financial markets. Investors need a reliable, data-driven model that can learn from historical trends and assist in forecasting near-future stock prices with reasonable accuracy.



High Volatility

Prices fluctuate rapidly due to news, sentiment and global events.



Non-linear Patterns

Simple linear models fail to capture complex market behaviour.



Risk of Loss

Poor predictions can lead to significant financial losses for investors.



Many Variables

Price depends on volume, sentiment, macro factors & company news.

OBJECTIVES

Project Objectives

1

Collect and pre-process historical stock market data (OHLC + Volume).

2

Perform exploratory data analysis to identify trends, seasonality and patterns.

3

Build and train ML / deep-learning models (e.g. LSTM, Linear Regression) to forecast prices.

4

Evaluate model performance using standard error metrics (RMSE, MAE, MAPE).

5

Visualize predicted vs. actual prices to assess model reliability.

6

Provide insights that can support smarter, data-backed investment decisions.

Existing Approaches to Price Forecasting

APPROACH	DESCRIPTION	LIMITATION
Fundamental Analysis	Evaluates a company's financial health, earnings, and industry position.	<i>Long-term, but slow to react to sudden change</i>
Technical Analysis	Uses historical price/volume charts and indicators (RSI, MACD, moving averages).	<i>Depends heavily on analyst interpretation</i>
Statistical Models (ARIMA)	Classical time-series models that assume linear relationships in data.	<i>Struggles with non-linear market behaviour</i>
Machine Learning (Our Focus)	Learns non-linear patterns directly from data using models like LSTM & Regression.	<i>Requires quality data & careful tuning</i>

Dataset Description

Historical stock data is typically sourced from public financial APIs and contains daily Open–High–Low–Close (OHLC) prices along with traded volume over several years.

Date	Trading day timestamp
-------------	-----------------------

Open	Price at market open
-------------	----------------------

High / Low	Day's price range
-------------------	-------------------

Close	Price at market close (target)
--------------	--------------------------------

Volume	Number of shares traded
---------------	-------------------------

COMMON DATA SOURCES

- Yahoo Finance API
- NSE / BSE Historical Data
- Alpha Vantage / Kaggle Datasets

5+ yrs

Historical range

70/30

Train / Test split

Data is cleaned, checked for missing values, and normalized (MinMax scaling) before model training.

Tools & Technologies Used



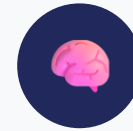
Python

Core programming language



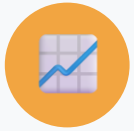
Pandas / NumPy

Data handling & processing



TensorFlow / Keras

Building the LSTM model



Matplotlib / Seaborn

Data visualization



Scikit-learn

Regression models & metrics



Jupyter Notebook

Development environment

System Architecture / Workflow

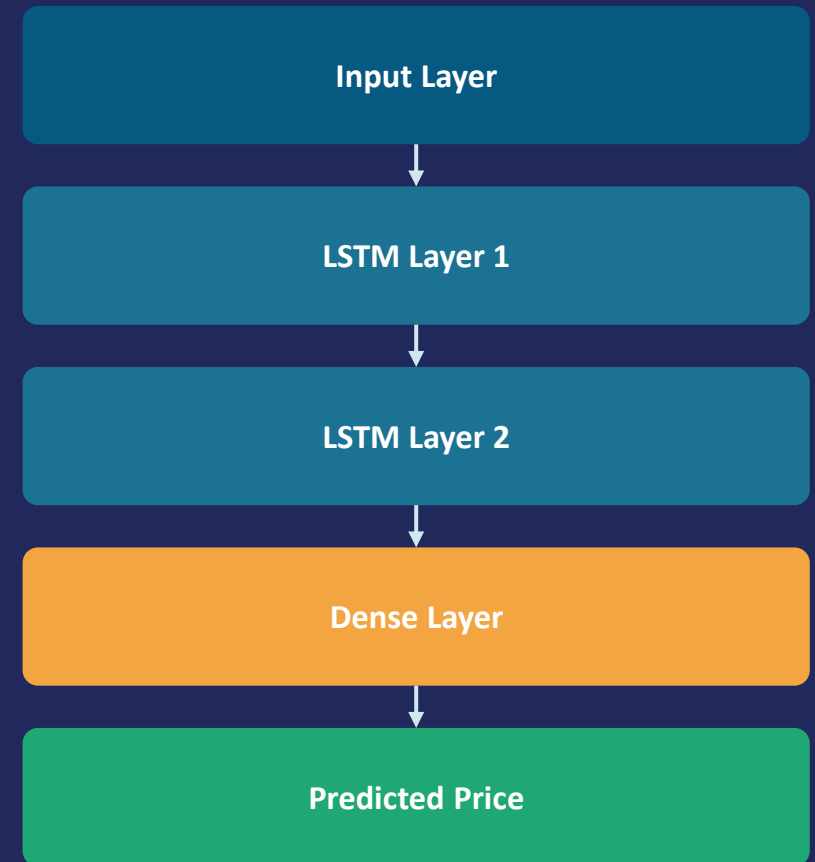


Raw market data flows through cleaning and feature engineering, is used to train a predictive model, and the model's output is evaluated and visualized against actual prices.

Why LSTM for Stock Prediction?

Long Short-Term Memory (LSTM) is a type of Recurrent Neural Network (RNN) specifically designed to learn patterns in sequential, time-dependent data — making it well suited to stock price series.

- ✓ Remembers long-term dependencies in sequential data
- ✓ Overcomes the vanishing-gradient problem of standard RNNs
- ✓ Uses gates (input, forget, output) to control information flow
- ✓ Well suited to noisy, time-ordered financial data



Step-by-Step Implementation

1

Import & Clean Data

Load historical stock data; handle missing values and outliers.

2

Normalize Data

Scale prices using MinMaxScaler for stable model training.

3

Create Sequences

Convert data into fixed-length time-window sequences (X, y).

4

Split Data

Divide into training and testing sets (e.g. 70% / 30%).

5

Build & Train Model

Define LSTM architecture and train on the training sequences.

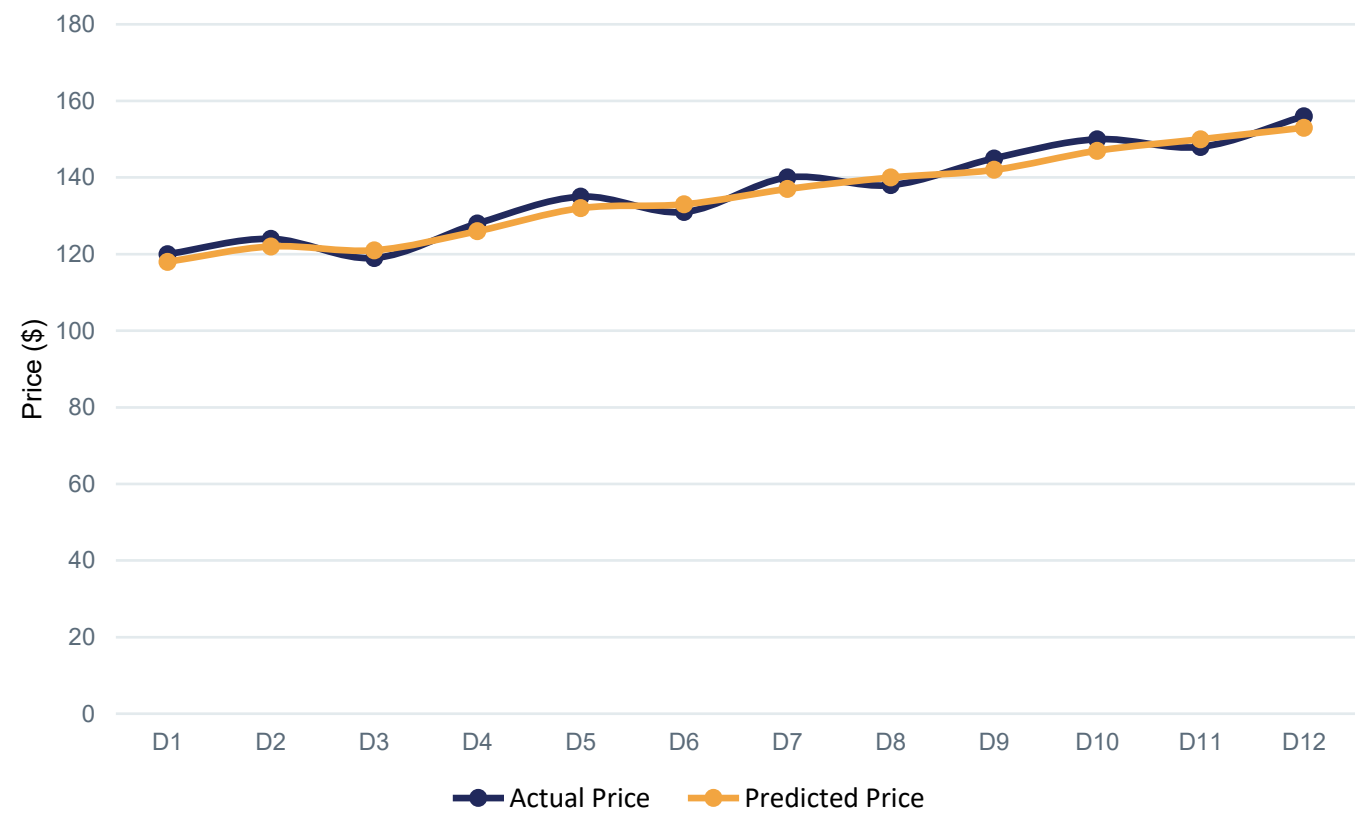
6

Predict & Evaluate

Forecast test prices and evaluate using RMSE / MAE / MAPE.

RESULTS

Actual vs. Predicted Prices (Sample)



3.42

RMSE

2.87

MAE

2.1%

MAPE

Lower RMSE / MAE / MAPE values indicate the model's predictions are closer to actual market prices.

Advantages & Real-World Applications

BENEFITS



Faster Decision-Making

Automated forecasts save analysts hours of manual chart study.



Pattern Detection

Identifies subtle trends invisible to manual analysis.



Scalable

Same model architecture can be applied across many stocks.

USE CASES



Algorithmic Trading

Powers automated trading bots and strategies.



Portfolio Management

Assists in risk assessment and asset allocation.



Investor Insight Tools

Feeds dashboards and apps used by retail investors.

Challenges & What's Next

CURRENT LIMITATIONS

- ✗ Cannot predict sudden market shocks or unexpected news events
- ✗ Model accuracy depends heavily on data quality and quantity
- ✗ Does not account for investor sentiment or macroeconomic shifts

FUTURE ENHANCEMENTS

- ✓ Incorporate sentiment analysis from news & social media
- ✓ Combine multiple models (ensemble/hybrid approach)
- ✓ Add real-time data streaming for live predictions

Conclusion

Stock price prediction using machine learning demonstrates how historical data and models like LSTM can uncover meaningful patterns in financial time series. While no model can perfectly predict the market, this project shows a practical, data-driven approach that can meaningfully support — not replace — human decision-making in investing.

KEY TAKEAWAYS

- ML models can learn non-linear market patterns
- LSTM handles sequential data effectively
- Predictions support — don't replace — human judgment
- Model performance improves with more/better data

Thank You

Questions & Discussion

Prince Kumar | Roll No. 25SCS1003002796 | IILM University, Greater Noida | Semester 3